

Mingjia Yang

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Education

University of Michigan , Ann Arbor, MI Ph.D. in Biophysics Advanced to candidacy, 2026	08/2025 – Present
Nanyang Technological University , Singapore B.Eng. in Bioengineering (Honours, Highest Distinction) Minor in Modern Languages	08/2021 – 05/2025

Research Interests

Single-molecule and super-resolution microscopy; quantitative live-cell imaging; computational image analysis for cellular dynamics and structure.

Research Experience

Heemskerk Lab , University of Michigan <i>Summer Research Project</i> Advisor: Assoc. Prof. Idse Heemskerk • Investigated how FGF/ERK signaling integrates with the BMP, WNT, and NODAL hierarchy in a human stem-cell model of gastrulation.	05/2026 – 09/2026
Joglekar Lab , University of Michigan <i>Rotation Student (half rotation)</i> Advisor: Prof. Ajit Joglekar • Built an image-analysis pipeline for tip-tracking EB3 comets at growing microtubule ends.	03/2026 – 04/2026
Heemskerk Lab , University of Michigan <i>Rotation Student (half rotation)</i> Advisor: Assoc. Prof. Idse Heemskerk • Contributed to characterizing a distinct developmental route from amnion to extra-embryonic mesoderm in a human stem-cell model.	01/2026 – 02/2026
Cai Lab , University of Michigan <i>Rotation Student</i> Advisor: Assoc. Prof. Dawen Cai • Developed a 3D single-molecule detection pipeline and the accompanying experimental protocol for connectome data from intact mouse brain.	09/2025 – 12/2025
Institute for Digital Molecular Analytics and Science (IDMxS) , Singapore <i>Final Year Project</i> Advisors: Asst. Prof. Matthew Foreman, Dr. Xiao-Liu Chu • Applied interference (iSCAT-based) imaging to structural analysis of SARS-CoV-2 virions.	09/2024 – 05/2025
Huang Group , Purdue University, USA <i>Visiting Research Intern</i> Advisor: Assoc. Prof. Fang Huang • Applied a Detection Transformer (DETR) architecture to single-molecule localization microscopy data. • Implemented preprocessing, filtering, and segmentation for super-resolution image stacks.	07/2024 – 09/2024
Institute for Digital Molecular Analytics and Science (IDMxS) , Singapore <i>Undergraduate Researcher (URECA)</i> Advisors: Asst. Prof. Matthew Foreman, Dr. Xiao-Liu Chu • Engineered nanoparticle–DNA oligonucleotide conjugates for an ultrasensitive detection assay. • Built a compact smartphone-attachable microscope for low-cost detection of small molecules and pathogens. • Designed an automated acquisition setup that increased imaging throughput fivefold. • Processed images in MATLAB and trained a CNN classifier in PyTorch to 98% accuracy.	08/2023 – 06/2024
Singapore Eye Research Institute (SERI) , Singapore <i>Data Science / AI Intern, Ophthalmology Group</i> Advisor: Asst. Prof. Yih Chung Tham • Curated a benchmark ophthalmic question-answering dataset for evaluating large language models. • Contributed to data analysis, manuscript preparation, and a funding proposal.	01/2023 – 05/2024
Tong Ling Lab , Nanyang Technological University, Singapore <i>Part-Time Research Assistant</i> Advisor: Asst. Prof. Tong Ling • Maintained and aligned optical systems and performed image processing for single-molecule detection on an iSCAT setup.	09/2023 – 12/2023
Merck Sharp & Dohme (MSD) , Singapore <i>Research Intern, Biomarker & Target Sciences</i>	05/2023 – 08/2023

- Optimized an *in vitro* idiopathic pulmonary fibrosis model, determining seeding density and serine protease concentrations that induce fibroblast contractility.
- Established model readouts by flow cytometry and qPCR, profiling canonical marker expression and cytokine-driven changes in fibroblast state.

Tong Ling Lab, Nanyang Technological University, Singapore

08/2022 – 06/2023

Undergraduate Researcher | Advisor: Asst. Prof. Tong Ling

- Implemented iSCAT microscopy to track live-cell membrane dynamics with nanoscale precision.
- Built iSCAT systems from the ground up, resolved optical alignment faults, and identified single-molecule motion using gold nanoparticle labels.
- Processed and tracked nanoparticle trajectories in MATLAB and Python; compared intensity fluctuations and power spectral densities across components and interpreted their biological implications.

Publications

- [1] S. Srinivasan, X. Ai, T.W.S. Lo, A. Gilson, M. Zou, K. Zou, H. Kim, **M. Yang**, K. Pushpanathan, S. Yew, W.T. Loke, J. Goh, Y. Chen, Y. Kong, E.Y. Fu, M.O. Hui, K. Nwanyanwu, A. Dave, K.Z. Li, C.-H. Sun, M. Chia, G.D. Yang, W.M. Wong, D.Z. Chen, D. Liu, M. Singer, F. Antaki, L.V. Del Priore, J. Jonas, R. Adelman, Q. Chen, Y.-C. Tham. Benchmarking LLMs for Ophthalmology (BELO) for ophthalmological knowledge and reasoning. *Ophthalmology Science* 6(3), 101050 (2026). doi:10.1016/j.xops.2025.101050

Conference Presentations

International Conference of Undergraduate Research (ICUR) 06/2024

- Poster: *Cost-Effective Nanoparticle-Based Assay with Automated Imaging for Ultrasensitive DNA Detection.*

URECA Poster Competition, Nanyang Technological University 03/2024

- Poster: *Nano-Imaging: Detecting Diseases Affordably.*

URECA Poster Competition, Nanyang Technological University 03/2023

- Poster: *Super-Resolution Interferometric Scattering Microscopy in Live Cells.*

Awards and Scholarships

Dean's List, Nanyang Technological University 2023 – 2025

- Awarded annually to the top 5% of the cohort.

NTU President Research Scholar 2022 – 2024

- Awarded for research accomplishment with Distinction in the URECA undergraduate research program.

Teaching Experience

Graduate Student Instructor, University of Michigan 01/2026 – 04/2026

Biophysics 290 — Physics of the Body and Mind

Peer Tutor, Nanyang Technological University 02/2025 – 05/2025

Bioelectricity; Signal Processing; Engineering Mathematics

Technical Skills

Experimental: iSCAT and interference microscopy, single-molecule localization microscopy, optical system design and alignment, mammalian and stem cell culture, immunofluorescence, flow cytometry, qPCR, nanoparticle functionalization, PCB design and printing

Computational: MATLAB, Python (PyTorch, NumPy, scikit-image), R, front-end web development; image segmentation and single-particle tracking; deep learning for image analysis

Software: ImageJ/Fiji, PyMOL, Prism, Figma, Fusion 360, Creo Parametric, MS Office

Languages: Mandarin (native), English (fluent), German (CEFR B1)

References

Dr. Fang Huang fanghuang@purdue.edu

Reilly Associate Professor, Department of Biomedical Engineering
Purdue University

Dr. Matthew Foreman matthew.foreman@ntu.edu.sg

Assistant Professor, School of Electrical & Electronic Eng.
Nanyang Technological University

Dr. Yih Chung Tham thamyc@nus.edu.sg

Assistant Professor, Yong Loo Lin School of Medicine
National University of Singapore